ATCM: Aerial–Terrestrial LiDAR-Based Collaborative Simultaneous Localization and Mapping

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***Abstract*— Multi-robot collaborative simultaneous localization and mapping (C-SLAM) offers precise scene reconstruction over single-robot SLAM and enables data fusion from heterogeneous robots. However, heterogeneous C-SLAM faces challenges in both accurate inter-robot loop closure detection and globally consistent data fusion due to inherent viewpoint disparities and heterogeneous data characteristics. This article introduces ATCM, an aerial–terrestrial light detection and ranging (LiDAR)-based C-SLAM method designed for heterogeneous robots without priori initial relative position. ATCM comprises three modules: single-robot front end employing diverse odometry methods, multi-robot loop closure detection, and global pose graph optimization (GPGO). A novel LiDAR-based cross-view global loop descriptor (CGLD) is proposed for scan-to-scan heterogeneous inter-robot loop closure detection. By uniformly mapping cross-view information into the height domain and integrating dynamic height, the loop descriptor automatically achieves viewpoint correction. In addition, we introduce a bidirectional loop detection (BLD) algorithm that validates inter-robot loop closures through both forward and reverse detections. Finally, the two-stage GPGO integrates multisource measurements, ensuring globally consistent mapping and localization with cross-view data. We have validated the effectiveness of ATCM on campus scenario datasets and the KITTI dataset, achieving a remarkable 21.95% improvement in trajectory accuracy and a 17.00% enhancement in map precision compared to high-precision point cloud maps, surpassing state-of-the-art LiDAR-based odometry methods. In the ablation experiments, the proposed loop descriptor achieved 97% accuracy in recognizing heterogeneous inter-robot loop closures. Moreover, compared to the traditional unidirectional method, the BLD method demonstrates up to a 31.2% improvement in loop closure accuracy.**

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I. Introduction

POINT clouds are increasingly introduced to digitally represent intricate environments from the physical world [1]. In recent years, the diversity of 3-D light detection and ranging (LiDAR)-based laser scanning systems has expanded significantly [2], [3], [4], enabling the rapid acquisition of high-quality point clouds through the integration of simultaneous localization and mapping (SLAM) technology and providing a solid data foundation for subsequent point cloud scene understanding [5] and recognition [6]. Substantial research has been devoted to single-robot SLAM, showcasing notable advancements in multisensor fusion SLAM [7], [8], [9], [10] and calibration [11], [12].

Despite significant progress, single-robot SLAM exhibits limitations including cumulative errors and inefficiency in large-scale and complex environments. Consequently, this has given rise to multi-robot collaborative SLAM (C-SLAM) as an extension of single-robot SLAM [13]. C-SLAM offers the advantages of effective and comprehensive reconstruction and has been applied in diverse scenarios, including underground spaces [14]. If the initial relative poses between robots are unknown, the direct fusion of multiple robots' maps or rigid registration based on overlapping regions may introduce inconsistencies in the global fusion map, as shown in Fig. 1. To overcome these limitations, C-SLAM effectively reduces the errors from single-robot SLAM and enables globally consistent data fusion even without prior knowledge of the robots' relative positions. C-SLAM comprises two main modules: the front end and the back end. The front end acquires intra-robot and inter-robot loop closures, along with landmark estimations and odometry constraints. Precise loop closure is essential for aligning robot poses within a common frame, typically achieved through place recognition (PR), which serves as the first key aspect in the fusion process. The back end utilizes multisource constraints to compute globally consistent poses and maps, representing the second key aspect in the

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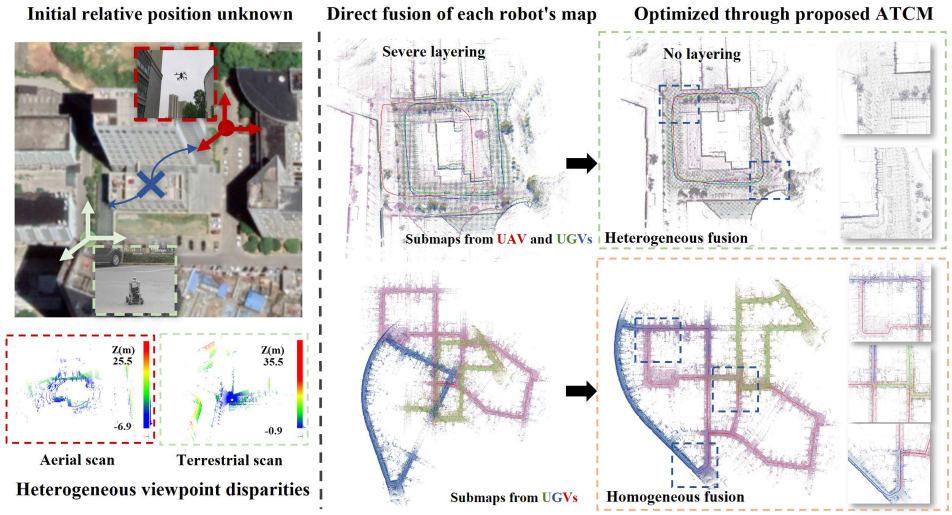


Fig. 1. Comparison of global fusion maps: ATCM method and direct fusion. First, due to the unknown initial relative poses between robots, severe layering occurs when directly merging multiple submaps. Second, variations in viewpoints and LiDAR scanning characteristics between heterogeneous platforms lead to differences in point cloud distributions. Specifically, aerial scans focus on high-altitude features, while terrestrial scans capture dense ground-level geometric details. Aerial scans are acquired by UAV, while terrestrial scans are captured by UGV. These inconsistencies make it challenging to generate unified point cloud descriptors for heterogeneous loop closure detection. The proposed ATCM method effectively achieves globally consistent fusion for both heterogeneous and homogeneous data. Different colors indicate the maps and corresponding trajectories of individual robots.

fusion process. C-SLAM can be classified as homogeneous and heterogeneous depending on the robot configurations [15].

While several existing homogeneous C-SLAM [16], [17], [18] methods employ loop closure detection approaches [19], [20], [21], they rely on similar robot types with a singular perspective and limited information acquisition. In contrast, heterogeneous C-SLAM offers a more diverse perspective to enhance environmental representation [22] and uses the specific advantages of different robots for increased versatility and flexibility [23]. However, the inherent viewpoint disparities in aerial–terrestrial C-SLAM pose significant challenges to heterogeneous inter-robot loop closure detection [24], leading to difficulties in achieving globally consistent data fusion, as shown in Fig. 1. For heterogeneous C-SLAM methods, He et al. [22] achieves inter-robot loop closure detection through the extraction and correlation of obstacle contours, but it results in a limited number of loop closures and is less applicable in scenarios with few obstacles. Yue et al. [15] employ an information gain mechanism to initiate the fusion of multiresolution hybrid maps. The Stronger Together method [25] employs the viewpoint invariance of semantic information to fuse cross-view information. However, reliance on UAVs with cameras limits these methods in low-light conditions, while 3-D LiDAR offers superior robustness in challenging environments. Camera-based loop closure detection approaches cannot be seamlessly adapted to LiDAR-based methods. Moreover, current methods [21], [26] focus on homogeneous data with minimal viewpoint variations, which restricts their ability to perform cross-view loop closure detection. Significant challenges still persist in achieving accurate LiDAR-based loop closure detection for heterogeneous aerial–terrestrial robots.

In general, the heterogeneous C-SLAM faces three main challenges. The first challenge is that viewpoint disparities between aerial and terrestrial robots lead to differences in point cloud distributions, complicating heterogeneous inter-robot loop closure detection, as shown in Fig. 1. The second challenge lies in obtaining accurate inter-robot loop closure constraints for the back end, as coarse loop descriptors may lead to incorrect loop closures. The third challenge is to integrate various constraints to achieve globally consistent pose estimation and mapping. To address these challenges, we propose ATCM, an advanced heterogeneous aerial–terrestrial C-SLAM method that equips both aerial and terrestrial robots with 3-D LiDAR. ATCM comprises three core components: single-robot front end, multi-robot loop closure detection, and two-stage global pose graph optimization (GPGO). It decouples the front end from the back-end modules, establishing connections only through locally optimal poses and corresponding keyframes. We have validated ATCM on various datasets, including our in-house collected DCSI-ATCM dataset and the public KITTI dataset [27]. The key contributions are outlined as follows.

1) We propose ATCM, a heterogeneous aerial–terrestrial C-SLAM method compatible with various LiDAR SLAM algorithms. It eliminates the need for prior calibration of initial robot positions and enables the fusion of cross-view data from heterogeneous robots. Furthermore, ATCM employs two-stage GPGO using single-robot local poses and inter-robot loop constraints to achieve globally consistent mapping and localization.
2) A novel LiDAR-based cross-view global loop descriptor (CGLD) is introduced for detecting inter-robot loop closures between aerial and terrestrial robots. CGLD mitigates the impact of inconsistencies in the data distribution due to viewpoint disparities and automatically achieves viewpoint correction by integrating dynamic height information. In addition, the bidi-

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rectional loop detection (BLD) algorithm incorporates forward and reverse loop detection to validate loop closures, providing accurate loop constraints for subsequent optimization.

3) The effectiveness of the proposed ATCM is demonstrated on campus scenario datasets and the public KITTI dataset. The ATCM achieves significant improvements, including a 21.95% reduction in absolute pose error (APE) and a 17.00% enhancement in map precision, surpassing state-of-the-art LiDAR-based odometry methods. The CGLD attains 97% accuracy in detecting heterogeneous inter-robot loop closures, while the BLD algorithm demonstrates up to a 31.2% improvement in loop closure accuracy compared to traditional methods.

The organization of subsequent sections is given as follows. Section II reviews existing work in multi-robot C-SLAM and PR for robotics. Section III details the methods employed. Section IV showcases diverse experimental results and discussion, while Section V concludes with a summary.

II. Related Work

This section reviews related work, focusing on multi-robot SLAM and the challenges of PR in robotics tasks.

A. Collaborative SLAM

Presently, C-SLAM can be broadly categorized into homogenous and heterogeneous robot collaborations, with this section focusing on both types.

1) Homogeneous C-SLAM: For LiDAR-(inertial)-based methods, the online system for multi-robot SLAM with 3-D LiDARs [28] leverages the SegMatch algorithm [29] for loop closure detection and employs a sparse pose graph for the back end. LAMP2.0 [30] is a centralized C-SLAM system compatible with various input odometry methods and uses graph information, observability, and received signal strength indicators for inter-robot loop closure selection. In addition, the distributed DiSCo-SLAM framework [17] leverages scan context descriptors [20] for data exchange and introduces a two-stage global and local optimization method. Based on DiSCo-SLAM, the distributed DCL-SLAM framework [18] employs a flexible single-robot front end and LiDAR-Iris descriptors [21], proven effective in real-world applications.

For visual-(inertial)-based methods, Door-SLAM [16] is a fully distributed SLAM system relying on point-to-point communication. It includes a distributed pairwise consistent measurement algorithm for the pose graph and features a front end that utilizes NetVLAD for loop closure. Kimera-Multi [31] is the pioneering fully distributed SLAM system utilizing visual sensors to construct real-time 3-D mesh models with semantic labels. The centralized back-end system COVINS-G [32] is compatible with any VIO front end. It utilizes 2-D keypoint and pose fusion across multiple agents and employs BoW descriptors for PR. D2SLAM [33] is a VIO-based decentralized and distributed aerial swarm state estimation method that achieves high-precision and globally consistent state estimation.

For multisensors fusion methods, the distributed omnidirectional visual–inertial–UWB system omni-swarm [34] is designed to address the challenges of aerial swarm state estimation. Swarm-SLAM [35] is a decentralized multi-robot SLAM system supporting diverse sensors. It prioritizes loop closures among robots based on algebraic connectivity maximization.

Extensive research has focused on homogeneous C-SLAM, which integrates robots of the same type, leading to a relatively uniform perspective and limited information acquisition. In contrast, heterogeneous C-SLAM embraces diverse perspectives, proving essential for constructing a more comprehensive environmental representation.

2) Heterogeneous C-SLAM: Zhang and Singh [36] conducted experimental research on collaborative mapping between ground and aerial robots. Miller et al. [25] proposed Stronger Together, a heterogeneous multi-robot system that utilizes real-time semantic maps created by drones for localization, planning, and navigation. One limitation is that the UAV and UGV must initialize within the same mapped region. Yue et al. [15] presented an aerial-ground collaborative mapping framework based on a continuous-discrete model. Real-time local mapping is achieved in continuous space, and local maps are fused based on information gained in discrete space. Wang et al. [37] proposed a collaborative aerial-ground continuous risk mapping framework, which utilizes risk values for point cloud segmentation and establishes residuals for accurate relative transformations. He et al. [22] introduced a method for collaborative ground and aerial mapping, including robust odometry estimation and obstacle data association with neural networks.

Existing methods for heterogeneous robot collaborative mapping commonly involve UAVs equipped with cameras and UGVs equipped with 3-D LiDAR sensors. Moreover, loop closure detection relies on calculating the similarity in the spatial distribution of point clouds to identify potential loop closures. The inherent viewpoint disparities within heterogeneous robotic systems result in UAV scans focusing on higher-altitude areas, while UGV scans capture more detailed ground-level information. These distribution inconsistencies make it difficult to generate unified descriptors for heterogeneous loop closure detection. Hence, this article explores heterogeneous C-SLAM when both UAVs and UGVs are equipped with 3-D LiDAR sensors simultaneously.

B. Place Recognition for Robotics

In C-SLAM, the establishment of inter-robot loop closures acts as a pivotal link, integrating local poses from multiple robots into global poses [13]. In addition, in PR, the impact of viewpoint disparities on changes in point cloud distribution poses significant challenges for recognition capabilities. LiDAR-based methods remain unaffected by lighting changes, in contrast to vision-based PR methods. Therefore, this article investigates LiDAR-based PR methods, categorized into local descriptors utilizing point-level features, global descriptors that consider appearance, and learning-based approaches [18], [38].

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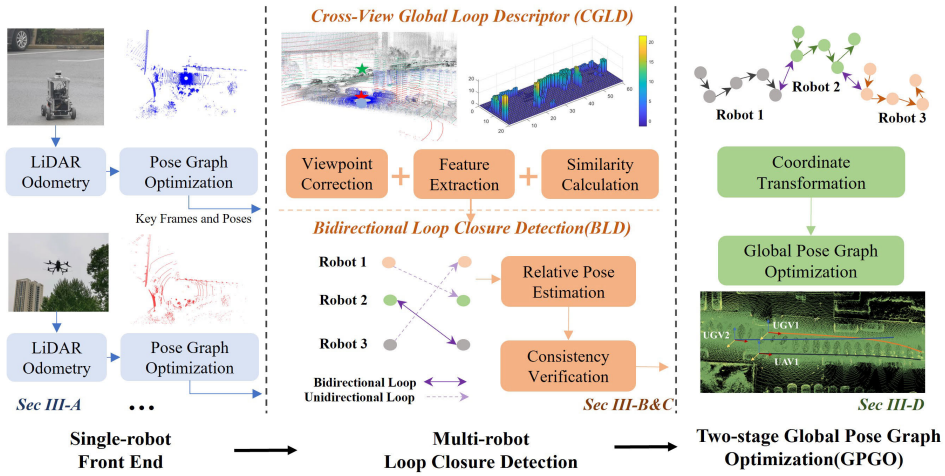


Fig. 2. Overview of the ATCM method includes three components: the single-robot front end, multi-robot loop closure detection, which incorporates the proposed CGLD and BLD algorithms, and the two-stage GPGO.

Local feature methods such as SHOT [39], often lack rotation and translation invariance, and are unsuitable for inter-robot PR applications because of their susceptibility to point cloud density and noise. For global descriptors, the rotation-invariant scan context [20] partitions point clouds in polar coordinates and encodes them through height projection, later refined to achieve both rotation and translation invariance [26]. Subsequently, Li et al. [40] integrate semantic information into the scan context, introducing a two-step global semantic ICP method for aligning point clouds. The intensity scan context descriptor [41] incorporates point cloud intensity and combines it with an efficient two-stage reidentification method. Xu et al. [42] proposed DiSCO, a LiDAR-based PR method leveraging frequency-domain transformation of features and utilizing spectrum magnitude to achieve rotation invariance.

However, existing 3-D LiDAR-based PR methods [20], [26], [43] are primarily tailored for homogeneous data and lack the ability to handle cross-view variations in the vertical dimension, particularly evident in heterogeneous aerial–terrestrial platforms. Approaches integrating semantic information and relying on learning-based techniques [44], [45], [46], [47] typically require pretraining and exhibit limited generalization across diverse scenes. Further verification of loop closures is essential to enhance the robustness, thereby improving the precision of loop closure constraints for PGO. Consequently, this article addresses the challenge of LiDAR-based inter-robot loop closure detection for aerial–terrestrial robots.

III. Methodology

The proposed ATCM framework for aerial–terrestrial robots, as depicted in Fig. 2, comprises three cores: single-robot front end, multi-robot loop closure detection, and two-stage GPGO. In the single-robot front-end module, environmental observation constraints are captured by each robot using a LiDAR sensor. Then, it employs various odometry methods, followed by keyframe selection and local PGO. The multi-robot loop closure detection integrates the local poses and keyframes to perform CGLD generation, BLD, relative pose estimation, and consistency verification. Subsequently, the back-end module utilizes the local poses of all robots and inter-robot loop constraints to perform the two-stage GPGO, solving for globally consistent optimal estimates of poses and maps.

A. Single-Robot Front End

The single-robot front end perceives the surrounding environment and estimates local poses using LiDAR-based odometry for each robot. To mitigate redundancy and reduce computation time, we implement a point cloud keyframe selection module. This module identifies the current frame as a keyframe if the cumulative pose change surpasses a predefined threshold κ from the previous state, as shown in Fig. 3. Subsequently, the local PGO module enhances local pose precision by integrating local odometry and loop closure measurements. The UGVs in the system are denoted as $\mathcal{R}_{\text{UGV}} = \{\beta, \eta, \ldots\}$ and the UAVs as $\mathcal{R}_{\text{UAV}} = \{\gamma, \delta, \ldots\}$, collectively denoted as $\mathcal{R} = \mathcal{R}_{\text{UGV}} \cup \mathcal{R}_{\text{UAV}}$ with n robots. For each robot $\forall \alpha \in \mathcal{R}$, its local pose graph $G^{L\alpha}$ includes the pose $p_{\alpha_i} \in \text{SE}(3)$ of node i within its local coordinate system. The pose p_{α_i} is further decomposed into $p_{\alpha_i} = (R_{\alpha_i}, t_{\alpha_i})$, where the rotation $R_{\alpha_i} \in \text{SO}(3)$ and the position $t_{\alpha_i} \in \mathbb{R}^3$. The ensemble of local poses for robot α is denoted as $P^{L\alpha} = \{p_{\alpha_1}, \ldots, p_{\alpha_i}\}$. As depicted in Fig. 3, local odometry measurements link successive robot poses, such as p_{α_i} and $p_{\alpha_{i+1}}$, while intra-robot loop measurements link nonsuccessive poses, such as $p_{\alpha_{i-1}}$ and $p_{\alpha_{i+1}}$. The measurement model for the relative pose p_{α_i} and p_{α_j} in $G^{L\alpha}$ is defined as follows:

$$\bar{z}_{\alpha_j}^{\alpha_i} \doteq \left(\bar{R}_{\alpha_j}^{\alpha_i}, \bar{t}_{\alpha_j}^{\alpha_i}\right), \quad i \neq j \tag{1}$$

$$\bar{R}_{\alpha_j}^{\alpha_i} = (R_{\alpha_i})^{\top} R_{\alpha_j} R_{\epsilon} \tag{2}$$

$$\bar{t}_{\alpha_j}^{\alpha_i} = (R_{\alpha_i})^{\text{T}}(t_{\alpha_j} - t_{\alpha_i}) + t_{\epsilon} \tag{3}$$

where $\bar{z}_{\alpha_j}^{\alpha_i}$ represents the relative pose measurement, incorporating both local odometry measurements and local intra-robot loop measurements. The relative rotation measurement $\bar{R}_{\alpha_j}^{\alpha_i}$ characterizes the attitude R_{α_j} in the node i reference frame

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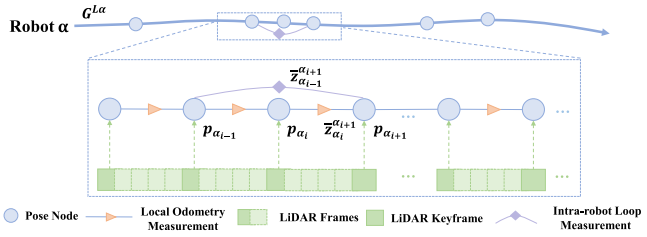


Fig. 3. Local pose graph for a single robot, including pose nodes, local odometry measurements, and intra-robot loop measurements.

with additional measurement noise rotation R_ϵ. Similarly, the relative position measurement $\bar{t}^{\alpha_i}_{\alpha_j}$ defines the position t_{α_j} relative to the reference frame at node i with added noise t_ϵ. $\xi^{L\alpha}_o$ and $\xi^{L\alpha}_l$ are the sets of local odometry and intra-robot loop measurements for robot α. In the local pose graph $G^{L\alpha}$, the set of all measurements is $\xi^{L\alpha} = \{\xi^{L\alpha}_l, \xi^{L\alpha}_o\}$. The maximum likelihood estimate for robot α can be expressed as the maximum likelihood of the measurement

$$P^{L\alpha *} = \arg\max_{P^{L\alpha}} \prod_{(\alpha_i,\alpha_j)\in\xi^{L\alpha}} \mathcal{L}\left(\bar{z}^{\alpha_i}_{\alpha_j} | P^{L\alpha}\right). \tag{4}$$

We solve (4) by methods [48], [49] to obtain the local optimal pose estimate for robot α, denoted as $P^{L\alpha *} = \{p^*_{\alpha_1}, \ldots, p^*_{\alpha_i}\}$ and the corresponding keyframe set $F^\alpha = \{F_{\alpha_1}, \ldots, F_{\alpha_i}\}$. Employing the same approach for other robots within the system, we acquire their respective sets of front-end local optimal pose estimates, denoted as $P^{L^*} = \{P^{L\alpha *}, P^{L\beta *}, \ldots, P^{L\gamma *}, P^{L\delta *}, \ldots\}$, along with the corresponding keyframes sets $F = \{F^\alpha, F^\beta, \ldots, F^\gamma, F^\delta, \ldots\}$.

B. Cross-View Global Loop Descriptor

Inter-robot loop closure detection is essential in scenarios with unknown initial relative poses among robots, using feature similarity for relative constraint estimation via point cloud matching. As shown in Fig. 4, the challenges for aerial–terrestrial loop closure detection are attributed to significant viewpoint differences between UAVs and UGVs, especially in vertical orientation. Inspired by Scan Context++ [26], we develop LiDAR-based CGLD with cross-view capabilities. Our approach differs in two key ways. First, by integrating dynamic height information, we can uniformly describe cross-view point clouds in the altitude domain, thereby avoiding categorical descriptions and mitigating data inconsistency caused by viewpoint disparities in heterogeneous robots. This enables efficient global feature representation, enhancing loop closure detection and integrating constraints into heterogeneous data fusion. Second, in the coarse-to-fine similarity calculation, we explicitly define proximity search spaces based on column resolution to reduce the impact of yaw angle discrepancies on loop closure, thereby improving robustness.

This section introduces point cloud feature description, focusing on CGLD generation, and coarse-to-fine similarity calculation to identify loop closures between corresponding keyframes.

1) Point Cloud Feature Description: For each robot $\alpha \in \mathcal{R}$, the keyframe from robot α is denoted as $F = \{f_1, f_2, \ldots, f_n\}$, $F \in F^\alpha$, and Z represents the elevation of the front-end odometry for this keyframe, which varies with changes in odometry. A point f_k in the point cloud can be defined in Cartesian coordinates as $f_k = \{x_k, y_k, z_k\}$ and in polar coordinates as $f_k = \{r_k, \theta_k, z_k\}$. By azimuthal and radial angle divisions in the sensor coordinate, the point cloud frame is described as N_s sectors and N_r rings, forming $N_r \times N_s$ bins, as illustrated in Fig. 4(b). The bin B_{ij} can be computed by the following equation, where $i \in \{1, 2, \ldots, N_s\}$ and $j \in \{1, 2, \ldots, N_r\}$. $L_{\max}$ represents the maximum perceptual distance of the LiDAR sensor

$$B_{ij} = \left\{ f_k \in F | \frac{(j-1)\cdot L_{\max}}{N_r} \le r_k < \frac{j\cdot L_{\max}}{N_r} \right. \\ \left. \frac{(i-1)\cdot 2\pi}{N_s} - \pi \le \theta_k < \frac{i\cdot 2\pi}{N_s} - \pi \right\}. \tag{5}$$

Each point is assigned to a bin based on its spatial coordinates. The viewpoint disparity at loop closures between UAVs and UGVs is prominently observed in the vertical dimension. To mitigate this disparity, an encoding function $\phi(\cdot)$, associated with the height domain, is employed for value assignment in (6). The function $\phi(\cdot)$ achieves viewpoint correction, enabling cross-view consistency in the vertical dimension

$$H_{ij} = \phi\left(B_{ij}\right) = \max_{f_k \in B_{ij}} (z(f_k) + h), \quad \text{where}$$
$$h = \begin{cases} Z, & \text{if } \alpha \in \mathcal{R}_{\text{UAV}} \\ \omega, & \text{if } \alpha \in \mathcal{R}_{\text{UGV}} \end{cases} \tag{6}$$

where $z(\cdot)$ retrieves the Z-value of point f_k. If bin B_{ij} has no points, H_{ij} is set to 0. For the UGV point cloud, no viewpoint correction is applied, i.e., $h = \omega$, where ω is the LiDAR sensor height for the UGV. For the UAV point cloud, the UAV takes off from the ground, and the front-end odometry algorithm continuously estimates the pose, thereby obtaining the dynamic odometry elevation Z. The CGLD D is accurately obtained using the following equation:

$$D = \left(H_{ij}\right) \in \mathbb{R}^{N_r \times N_s}. \tag{7}$$

2) Coarse-to-Fine Similarity Calculation: Describing point clouds is insufficient for determining loop closure and requires evaluating interframe similarity. In practical scenarios, yaw angle differences between robots relative to the global coordinate system may shift descriptor columns, potentially causing missed loop closures. Transforming descriptors in the column direction is essential, with sector keys introduced to accelerate this process, sharing properties with CGLD in Fig. 4(b) and (c). Given point cloud descriptors D_α and D_γ for robot α and γ, the corresponding sector keys are denoted as E_α and E_γ, as defined by the following equation:

$$E = (e_i) = \left(\frac{1}{N_r} \sum_{j=1}^{N_r} H_{ij} \right) \in \mathbb{R}^{1 \times N_s}. \tag{8}$$

Employing a columnwise transformation with a resolution of $(2\pi/N_s)$, the transformed descriptors of E_γ are computed,

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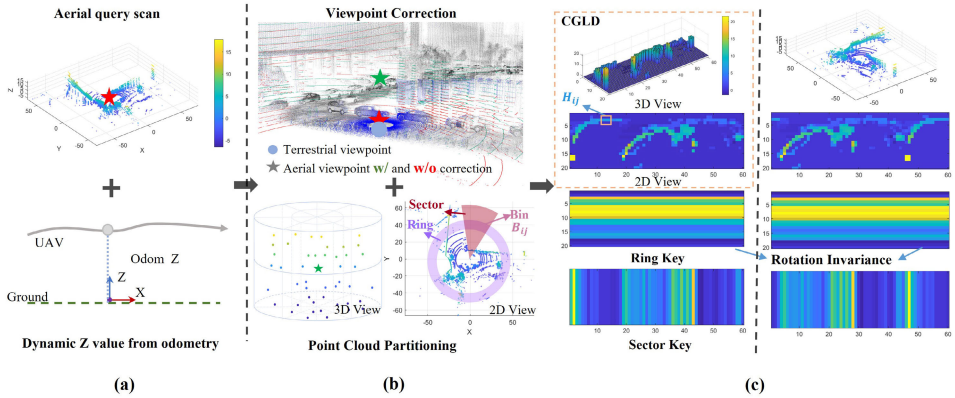


Fig. 4. CGLD generation workflow. During the Z-axis rotation of the point cloud, the ring key exhibits rotation invariance, while the sector key and CGLD undergo column transformations. (a) Data input. (b) Feature description (CGLD). (c) CGLD from the same scan after 90° Z-axis rotation.

represented as E_γ^i where $i \in \{1, 2, \ldots, N_s\}$. Subsequently, the following equation is applied to calculate the optimal transformation index $i_{\min}$:

$$i_{\min} = \arg\min_{i \in N_s} \left\| E_\alpha - E_\gamma^i \right\|. \tag{9}$$

To refine the robustness of the interframe similarity calculation despite the coarse description of the sector key, we employ the coarse-to-fine strategy, avoiding direct utilization of $i_{\min}$ to transform D_γ. First, using $i_{\min}$ as the central reference, the coarse transformation space K is computed according to the following equation, where ρ represents the ratio of searched columns to all columns:

$$K = \left\{ (i_{\min} + m + N_s) \mod N_s | m \in \left[-\left\lfloor \rho \cdot \frac{N_s}{2} \right\rfloor \right. \right. \left. \left. \left\lfloor \rho \cdot \frac{N_s}{2} \right\rfloor \right] \right\}. \tag{10}$$

Next, within the transformation space K, we perform columnwise transformations to generate a set of transformed descriptors for D_γ denoted as D_γ^k where $k \in K$. Subsequently, the minimum $s(D_\alpha, D_\gamma^k)$ is selected as the final similarity, as described in (11). The vectors c_l^α and c_l^γ represent column vectors at the same index in D_α and D_γ, respectively. Currently, by recording k, the yaw diff is calculated as the initial prior for relative pose estimation, contributing to an accelerated and accurate registration convergence. Moreover, in Fig. 5, we show the descriptor similarity between UAV and UGV using both dynamically height-corrected and uncorrected viewpoints

$$\begin{aligned} s(D_\alpha, D_\gamma) &= \frac{1}{N_s} \sum_{l=1}^{N_s} \left(1 - \frac{c_l^\alpha \cdot c_l^\gamma}{\|c_l^\alpha\| \|c_l^\gamma\|} \right) \\ S(D_\alpha, D_\gamma) &= \min_{k \in K} s\left(D_\alpha, D_\gamma^k\right). \end{aligned} \tag{11}$$

C. Bidirectional Loop Closure Detection

For global PGO, precise loop closure constraints are crucial for integrating heterogeneous data, but traditional unidirectional loop retrieval lacks robustness. To enhance accuracy,

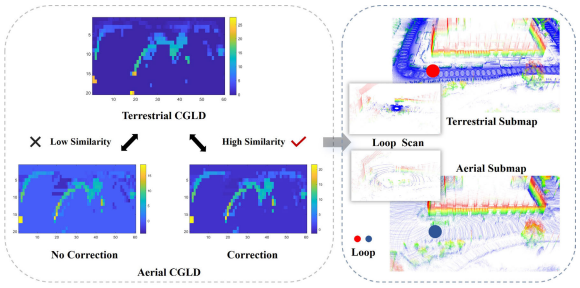


Fig. 5. Comparison of the effectiveness of aerial descriptors with and without viewpoint correction. If the viewpoint is not corrected, there is a low similarity between aerial descriptors and terrestrial descriptors, which are not deemed to constitute a loop.

we propose the BLD method for further validating the correctness of loop closures. The BLD algorithm is divided into two parts: forward loop detection and reverse loop detection. Subsequently, we compute relative pose constraints and validate the correctness of the detected loop closures through geometric consistency. $\forall \alpha \in \mathcal{R}$, leveraging Section III-B, keyframes F^α are transformed into $D^\alpha = \{D_{\alpha_1}, \ldots, D_{\alpha_i}\}$, where D_{α_i} signifies the descriptor of the ith keyframe of the robot α. To facilitate loop closure retrieval, we introduce a rotation-invariant descriptor ring key Q. This simplified representation not only mitigates viewpoint differences in the vertical dimension but also accommodates variations due to differing yaw angles, as illustrated in Fig. 4(b) and (c). Leveraging (12), we construct the ring key Q. Q_{α_i} represents the ring key of the ith keyframe of robot α

$$Q = (q_j) = \left(\frac{1}{N_s} \sum_{i=1}^{N_s} H_{ij} \right) \in \mathbb{R}^{N_r \times 1}. \tag{12}$$

In the forward loop detection between robots α and β where $\forall\, \alpha, \beta \in \mathcal{R}$, robot α serves as the target robot, while robot β is the source robot. The keyframe sets F^β of source robot β search for loop closures in the keyframe sets F^α of target robot α. Referring to Algorithm 1, a KD-tree $K^{\alpha\beta}$ is constructed using Q^α and Q^β to search loop closures with F^α as the reference. The loop correspondence $f^{\alpha\beta}$, representing the relationship between keyframe sets F^β and F^α, with F^α

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Algorithm 1 Forward Loop Closure Detection

Input: Keyframe sets F^{α} and F^{β} of target robot α and source robot β, loop similarity threshold t.
Output: Loop correspondence function $f^{\alpha\beta}$.
Initialize descriptor sets $D^{\alpha\beta}$, ring key sets $Q^{\alpha\beta}$ and KD-tree $K^{\alpha\beta}$ for robots α and β.
for each keyframe $F_{\alpha_m} \in F^{\alpha}$ and $F_{\beta_n} \in F^{\beta}$ **do**
  Calculate D_{α_m}, Q_{α_m} and D_{β_n}, Q_{β_n}.
  $D^{\alpha\beta} \leftarrow D_{\alpha_m}, D_{\beta_n}$.
  $Q^{\alpha\beta} \leftarrow Q_{\alpha_m}, Q_{\beta_n}$.
  $K^{\alpha\beta} \leftarrow Q^{\alpha\beta}$.
  if F_{β_n} **then**
    Traverse the index set using coarse-to-fine similarity calculation method to find $S_{\min}$ and its index.
    if $S_{\min} < t$ **then**
      update $f^{\alpha\beta}$.
    end if
  end if
end for
return $f^{\alpha\beta}$.

as the reference, is expressed as follows:

$$f^{\alpha\beta} : F^{\beta} \rightarrow F^{\alpha}. \tag{13}$$

In the reverse loop detection, robot α acts as the source robot and robot β as the target robot. Similar to Algorithm 1, the loop correspondence $f^{\beta\alpha}: F^{\alpha} \rightarrow F^{\beta}$ with F^{β} as the reference is obtained. To improve the precision of loop closures, the relationship between $f^{\alpha\beta}$ and $f^{\beta\alpha}$ is examined. Bidirectional loop closure is recognized through F_{β_n} and F_{α_m} when (14) is satisfied. This means that with F^{α} as the reference, the nth keyframe F_{β_n} of robot β matches the mth keyframe F_{α_m} of robot α as the loop closure. Similarly, with F^{β} as the reference, the mth keyframe F_{α_m} of robot α aligns with the nth keyframe F_{β_n} of robot β. Through bidirectional loop closure, the refined loop correspondence $f_b^{\alpha\beta}$ between robots α and β is determined

$$\begin{aligned} f^{\alpha\beta}\left(F_{\beta_n}\right) &= F_{\alpha_m} \\ f^{\beta\alpha}\left(f^{\alpha\beta}\left(F_{\beta_n}\right)\right) &= F_{\beta_n}. \end{aligned} \tag{14}$$

Relative Pose Estimation and Consistency Verification: After BLD, estimating relative pose constraints between robots becomes crucial based on the inter-robot loop correspondence $f_b^{\alpha\beta}$. Limitations in CGLD's representation may lead to inaccuracies in loop closures, requiring further refinement. To address these, we employ the Fast-GICP [50] algorithm to compute relative pose constraints and verify geometric consistency. The yaw angle differences obtained during loop retrieval serve as the initial prior for Fast-GICP. The loop closure is confirmed through geometric consistency validation if the fitness score falls below the predefined threshold φ, and its constraints are employed for subsequent back-end GPGO.

D. Two-Stage GPGO

The solution to the single-robot PGO problem can be expanded to multi-robot PGO. The two-stage GPGO

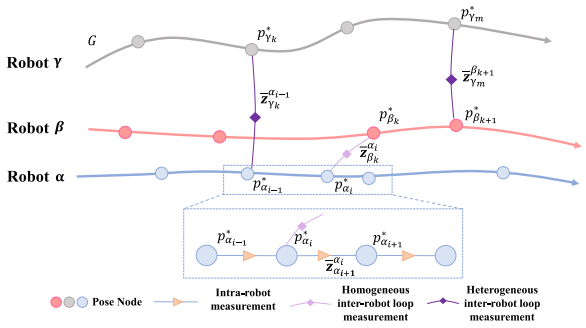


Fig. 6. Global pose graph, involving pose nodes, intra-robot measurements, homogeneous inter-robot, and heterogeneous inter-robot loop measurements.

collaboratively optimizes local poses and integrates all loop closure constraints, relying only on locally estimated poses and keyframes obtained from the front end to ensure decoupling. It comprises two phases: the coordinate transformation and the global PGO. The first-stage coordinate transformation provides initial values of nodes for the second-stage PGO and prevents local convergence during optimization. The method constructs the global pose graph G, which includes the local optimal pose set for all robots P^{L^*}, along with intra-robot and inter-robot measurements, as illustrated in Fig. 6. The inter-robot sloop closure measurement model for the relative pose between $p_{\alpha_i}^*$ and $p_{\beta_k}^*$ can be expressed

$$\begin{aligned} \bar{z}_{\beta_k}^{\alpha_i} &\doteq \left(\bar{R}_{\beta_k}^{\alpha_i}, \bar{t}_{\beta_k}^{\alpha_i}\right) \\ \bar{R}_{\beta_k}^{\alpha_i} &= \left(R_{\alpha_i}\right)^{\top} R_{\beta_k} R_{\epsilon} \\ \bar{t}_{\beta_k}^{\alpha_i} &= \left(R_{\alpha_i}\right)^{\mathrm{T}}\left(t_{\beta_k} - t_{\alpha_i}\right) + t_{\epsilon} \end{aligned} \tag{15}$$

where the relative rotation measurements $\bar{R}_{\beta_k}^{\alpha_i}$ and the relative position measurement $\bar{t}_{\beta_k}^{\alpha_i}$ describe the attitude R_{β_k} and the position t_{β_k} of robot β relative to node i of robot α in its own reference frame. The back-end PGO handles two types of inter-robot loop measurements: homogeneous measurements, represented as $\bar{z}_{\beta_k}^{\alpha_i}$ and $\bar{z}_{\delta_k}^{\gamma_i}$, and heterogeneous measurements, denoted as $\bar{z}_{\gamma_k}^{\alpha_i}$.

1) Coordinate Transformation: The local frame $\mathcal{F}_L^{\alpha}$ for the robot α is defined to align with its first frame $\mathcal{F}_1^{\alpha}$. Similarly, the global frame $\mathcal{F}^G$ is established in relation to $\mathcal{F}_L^{\alpha}$ of robot α. From the front end, we can determine the locally optimal pose $p_{\alpha_i}^*$ of the ith frame of robot α with respect to $\mathcal{F}_L^{\alpha}$ and $p_{\beta_k}^*$ of the kth frame of robot β relative to $\mathcal{F}_L^{\beta}$. Using the inter-robot loop measurement $\bar{z}_{\beta_k}^{\alpha_i}$, we can compute the transformation matrix T_{β}^{α} of the local frame $\mathcal{F}_L^{\beta}$ to $\mathcal{F}_L^{\alpha}$

$$T_{\beta}^{\alpha} = p_{\alpha_i}^* \cdot \bar{z}_{\beta_k}^{\alpha_i} \cdot \left(p_{\beta_k}^*\right)^{-1}. \tag{16}$$

If the ith point cloud of robot α along with the kth point cloud of robot β is the first reliable loop closure, we use the transformation matrix T_{β}^{α} to transfer the set of poses $\{p_{\beta_1}^*, \ldots, p_{\beta_k}^*\}$ from robot β to the local frame $\mathcal{F}_L^{\alpha}$ by $p_{\beta_k}^{\alpha} = T_{\beta}^{\alpha} \cdot p_{\beta_k}^*$. With a new reliable loop closure constraint $\bar{z}_{\beta_{k+m}}^{\alpha_{i+n}}$, we update T_{β}^{α} using (16) and transfer the set of poses $\{p_{\beta_{k+1}}^*, \ldots, p_{\beta_{k+m}}^*\}$ to the local frame $\mathcal{F}_L^{\alpha}$. This coordinate transformation aligns local frames to the global frame $\mathcal{F}^G$,

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providing the global pose graph G with initial values P of the nodes.

2) Global Pose Graph Optimization: The intra-robot measurements for robot α are denoted as ξ_I^α, and all intra-robot measurements in the system are denoted by $\xi_I = \{\xi_I^\alpha, \xi_I^\beta, \ldots, \xi_I^\gamma, \xi_I^\delta, \ldots\}$.Homogeneous and heterogeneous inter-robot loop measurements are designated as ξ_{ho} and ξ_{he}. The comprehensive set of all measurements is expressed as $\xi = \{\xi_I, \xi_{\text{ho}}, \xi_{\text{he}}\}$, and the set of inter-robot loop measurements is $\xi_h = \{\xi_{\text{ho}}, \xi_{\text{he}}\}$. By extending (4) for multi-robot maximum likelihood trajectory estimation, we can obtain

$$P^* = \arg\max_P \prod_{(\alpha_i,\alpha_j)\in\xi_I} \mathcal{L}\left(\bar{z}_{\alpha_j}^{\alpha_i} | P\right) \prod_{(\alpha_i,\beta_k)\in\xi_h} \mathcal{L}\left(\bar{z}_{\beta_k}^{\alpha_i} | P\right). \quad (17)$$

The globally optimal pose set P^* in the world frame $\mathcal{F}^G$ can be obtained. After GPGO, the local coordinate systems of all robots are uniformly aligned with the global coordinate system.

IV. Experiments

A. Dataset and Configuration

1) Dataset Introduction: In our experiments, we validate the algorithm using two datasets: 1) the in-house collected DCSI-ATCM dataset and 2) the publicly available KITTI00 dataset. Most existing multi-robot SLAM datasets [51], [52] focus on homogeneous data, lacking heterogeneous aerial–terrestrial data. Thus, we conducted experiments on DCSI-ATCM and KITTI datasets using a data segmentation approach, following prior works [17], [35].

a) DCSI-ATCM dataset: We employ a DJI M300 UAV equipped with a VLP16 LiDAR and a UGV fit with an Ouster OS1 128 LiDAR to collect data on campus, as shown in Fig. 1. Table I outlines the dataset parameters. We obtain highly accurate ground-truth trajectories for the UAV and UGV sequences using the CAOM [53] algorithm. However, these trajectories are collected in their respective local coordinate systems, and a priori relative positions of the UAV and UGV are unknown, resulting in the lack of a high-precision global reference trajectory for the UAV-UGV. To acquire a high-precision map for experimental accuracy assessment, we employ the RIEGL VZ-400i terrestrial laser scanning (TLS) to generate a comprehensive point cloud map.

b) Dataset partitioning and utilization: To simulate multi-robot data, we partitioned the datasets as detailed in Table I, and the resulting trajectories in Fig. 7 show overlaps that facilitate loop closures between robots. For homogeneous experiments, we use the partitioned KITTI00 and the UAV portion of the DCSI-ATCM dataset (referred to as the DCSI-UAV dataset), while for heterogeneous experiments, we select two UGVs and UAV2 from the partitioned DCSI-ATCM.

2) Parameter Setting and Implementation: The front-end algorithm in ATCM adopts the SOTA LiDAR odometry SCALOAM[1], CT-ICP [54], and KISS-ICP [55] in the experiment. Since SCALOAM incorporates an integrated single-robot PGO, we directly utilized the official implementation. For map accuracy comparison, we use the ground-truth map as the reference to evaluate the front-end global maps generated by different algorithms against our method. The front-end global map is created by manually registering local maps from each robot using overlapping regions. For the DCSI-ATCM dataset, CloudCompare[2] is used to align the UAV and UGV2 maps to the UGV1 map, and for the KITTI dataset, maps are referenced to the first robot. We then compute point-to-point distances between each map and the ground truth to assess quality after standardizing all maps to the same reference frame. This computation utilized $K = 6$ nearest neighbors for local point cloud fitting and the least square plane fitting mode within CloudCompare.

In the experimental setup, parameters N_s, N_r, and ρ of CGLD are configured as 60, 20, and 0.1 for all datasets. The keyframe parameter κ is set to 2.5 for the extensive KITTI dataset. Due to the sparse point clouds and instability of UAV data, κ is set to 0.5 for UAVs and 1 for UGVs. The similarity threshold t is set to 0.3 for KITTI and 0.2 for DCSI-UAV, while the geometric consistency threshold φ is 0.5 for both datasets. Taking into account the heterogeneous data discrepancy within the DCSI-ATCM dataset, the thresholds t and φ are elevated to 0.3 and 1.

The experiments utilized a computer featuring an Intel[3] i7-11800H @ 2.30-GHz CPU and 16 GB of RAM. The algorithm, implemented in C++, runs on the Ubuntu Linux 18.04 system, utilizing the robot operating system (ROS). The evaluation of trajectory accuracy in these experiments is conducted using the evo tool[4], with quantification based on the metrics of APE and relative pose error (RPE) [56]. We use the Umeyama algorithm [57] for alignment of the trajectory estimation with the ground-truth trajectory.

TABLE I
Detailed Information of Experiment Datasets

Dataset		LiDAR	Number of Frames
KITTI00 (R1/R2/R3)		Velodyne HDL-64E	4541 (2460/1091/990)
DCSI–ATCM	UGV (R1/R2/R3)	Ouster OS1 128	6874 (2368/2150/2356)
	UAV (R1/R2/R3)	Velodyne VLP-16	11560 (4363/3322/3875)

Note: R1, R2, and R3 represent different segments within each dataset for robots.

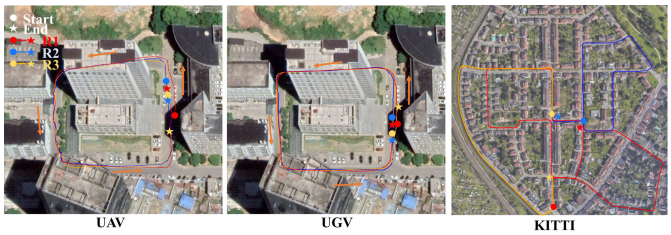


Fig. 7. Trajectory partitioning schematic for the DCSI-ATCM and KITTI datasets.

[1]https://github.com/gisbi-kim/SC-A-LOAM

[2]https://github.com/CloudCompare/CloudCompare

[3]Registered trademark.

[4]https://github.com/MichaelGrupp/evo

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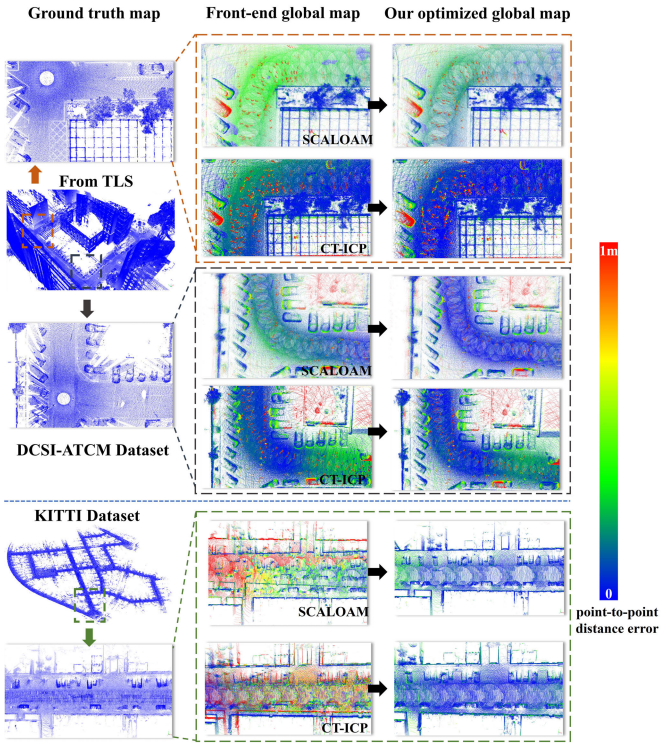
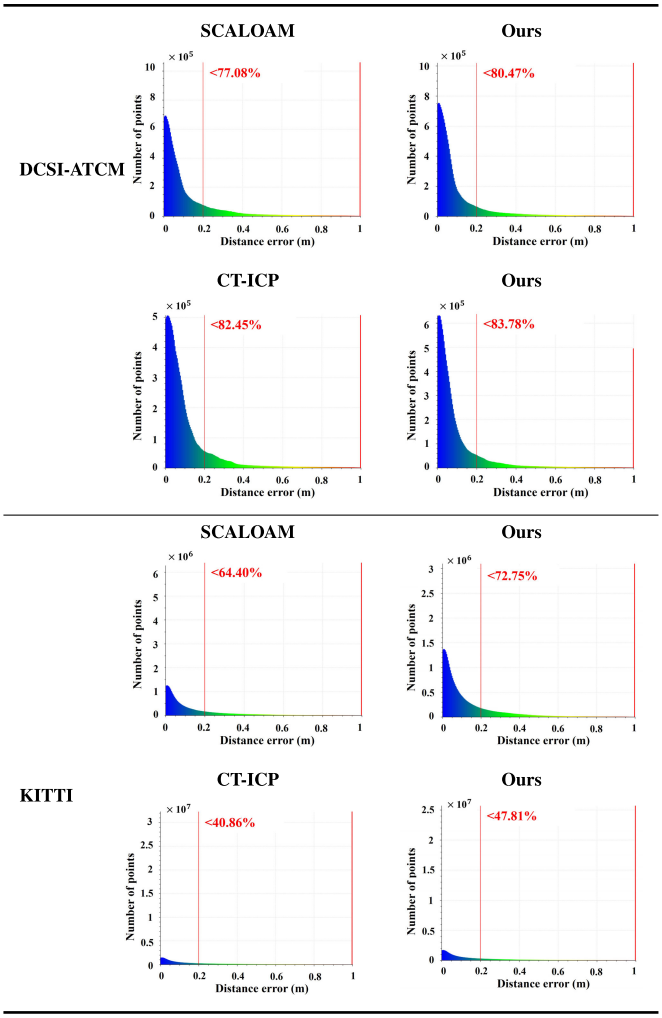


Fig. 8. Comparison of map accuracy between our global maps and front-end global maps relative to the ground-truth map by different algorithms.

TABLE II

HISTOGRAMS COMPARISON OF MAP DISTANCE ERROR BETWEEN OUR GLOBAL MAPS AND FRONTEND GLOBAL MAPS RELATIVE TO THE GROUND-TRUTH MAP BY DIFFERENT ALGORITHMS



B. Single-Robot and Multi-Robot SLAM Performance Comparison

This section demonstrates the effectiveness of the proposed ATCM in integrating data from multiple heterogeneous robots. It mitigates cumulative errors from the front end, even without prior relative positions, resulting in a globally consistent map. Furthermore, it shows advantages over advanced mapping algorithms and rigid registration-based fusion methods, evaluated in terms of map accuracy and trajectory precision.

1) Map Accuracy Comparison: Due to severe layering issues in directly merging multiple robots' maps, we compare the global map generated by the ATCM method with the front-end global maps obtained through rigid registration on the KITTI and DCSI-ATCM datasets. The method for acquiring the front-end global maps is detailed in Section IV-A2. Fig. 8 illustrates point-to-point distance error map comparisons between our global maps and the front-end global maps relative to the ground-truth map, where colors indicate error size. In the DCSI-ATCM dataset, our method effectively integrates heterogeneous cross-view data from UAVs and UGVs, and there is no layering of building structures and vehicles. Areas with significant errors typically lack corresponding points in the ground-truth map, which are mainly dynamic pedestrians and changed vehicles and areas that cannot be collected by TLS. The differences between the point clouds are caused by the different collection times of the ground-truth map and DCSI-ATCM data, as well as the higher UAV perspective of DCSI-ATCM. In addition, our global map shows smaller ground errors compared to front-end global maps generated by SCALOAM and CT-ICP, approaching zero. In the KITTI dataset, our method integrates data from multiple UGVs without any prior relative positions. Despite the expansive scene scope leading to higher overall map errors, our map closely matches the ground truth at both the initial and final points without layering issues, achieving minimal distance errors compared to the front-end global maps from SCALOAM and CT-ICP.

Moreover, to precisely assess the accuracy of the maps, we compute histograms of map distance error while filtering out values exceeding 1 m. The histogram serves to sort distances from low to high and record the corresponding point counts. In Table II, we use the ratio of points with point-to-point distance error below the specified threshold to the total number of map points as our evaluation metric. For point-to-point distance error less than 0.2 m, our method exhibits map point percentages of 80.47% and 83.78% in the DCSI-ATCM dataset, corresponding to improvements of 4.39% and 1.61% over SCALOAM and CT-ICP. In the KITTI dataset, our method achieves map point percentages of 72.75%

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TABLE III
COMPARISON OF APE METRICS BETWEEN THE PROPOSED METHOD ATCM AND OTHER METHODS

Dataset	Robot	SCALOAM/ours		KISS-ICP/ours		CT-ICP/ours	
		RMSE	median	RMSE	median	RMSE	median
DCSI – UAV	UAV1	0.35/0.35	0.25/0.25	failed(64%)		failed(89%)	
	UAV2	0.30/0.30	0.15/**0.13**	failed(49%)		0.26/-	0.14/-
	UAV3	0.29/0.30	0.14/0.15	failed(63%)		failed(90%)	
	all	0.41/**0.32**	0.30/**0.16**	failed(24%)		failed(34%)	
KITTI	UGV1	1.00/1.01	0.81/0.88	4.39/**1.44**	4.31/**1.14**	3.17/**2.02**	2.78/**1.70**
	UGV2	0.58/**0.49**	0.48/**0.42**	0.74/**0.55**	0.68/**0.49**	0.58/0.58	0.56/0.56
	UGV3	1.71/**0.71**	1.21/**0.67**	1.35/**0.74**	0.98/**0.66**	1.53/**1.04**	1.37/**0.92**
	all	1.28/**1.07**	1.06/**0.85**	1.85/**1.70**	1.52/**1.46**	2.38/**2.33**	1.90/**1.82**

Note: Bold indicates superior metrics."ours' represents the proposed ATCM applied with different front-end algorithms, while 'all' represents the results from corresponding front-end algorithms using fully unsegmented datasets and ATCM. "%" indicates the percentage of time when the algorithm shows significant drift. Please refer to Section IV-A1 for further details about DCSI-UAV.

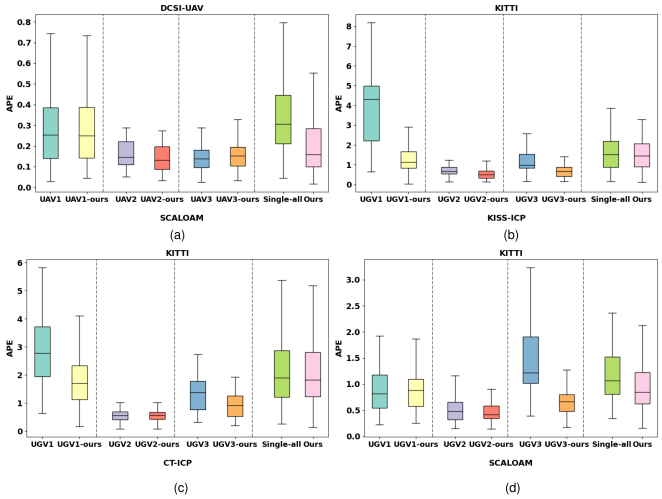


Fig. 9. Boxplot comparison of APE metrics between the proposed and other methods. (a) Based on SCALOAM in DCSI-UAV. (b) Based on KISS-ICP in KITTI. (c) Based on CT-ICP in KITTI. (d) Based on SCALOAM in KITTI.

and 47.81%, surpassing SCALOAM and CT-ICP by 12.96% and 17.00%, respectively. These results emphasize the superior accuracy and robustness of our algorithm across heterogeneous platforms.

2) Trajectory Accuracy Comparison: This section provides a quantitative demonstration of the superior trajectory accuracy achieved by the proposed ATCM method compared to current state-of-the-art LiDAR odometry methods. In addition, it effectively reduces the drift in front-end trajectories on the KITTI and DCSI-UAV datasets (refer to Section IV-A1 for details about DCSI-UAV). Table III and Fig. 9 present the APE comparison of our method with different approaches and datasets. First, we compare the local trajectory accuracy between ours and that achieved by the front-end algorithm, demonstrating our method's effectiveness in reducing front-end trajectory drift. Second, we compare the global trajectory accuracy between ours and that achieved by the front-end algorithm using the unsegmented dataset. Our algorithm not only reduces front-end drift but also outperforms the front-end algorithms on unsegmented datasets.

In the DCSI-UAV dataset, our algorithm improves global trajectory accuracy. Due to significant flight jitter causing data instability and frame matching difficulties, both KISS-ICP and CT-ICP fail on all UAVs except UAV2, where CT-ICP performs adequately. The percentage of time during which the algorithms exhibited large drift is recorded in Table III. For accuracy evaluation, we compared the RMSE of APE. For the global trajectory, SCALOAM achieves 0.41, whereas our method achieves 0.32, marking an improvement of 21.95%. In the KITTI dataset, our algorithm also significantly improves trajectory accuracy, particularly for robots with long routes and limited loop closures. For UGV1, our method achieves 1.44 and 2.02, representing enhancements of 67.19% and 36.27% over KISS-ICP and CT-ICP, respectively. Similarly, for UGV3, our method achieves 0.71 and 0.74, with improvements of 58.47% and 45.18% over SCALOAM and CT-ICP. In addition, our method achieves a global trajectory error of 1.07, outperforming SCALOAM by 16.40%.

C. Ablation Studies

1) Bidirectional Loop Closure Detection: This section evaluates the necessity of the proposed BLD on the DCSI-ATCM, DCSI-UAV, and KITTI datasets. We conduct a comparative assessment of BLD against the unidirectional loop closure detection method in terms of both loop closure accuracy and trajectory precision, using SCALOAM as the front-end method. The unidirectional loop closure detection method includes forward loop closure detection and consistency verification, denoted as FD + CV in Table IV. Our comparative analysis demonstrates that the BLD algorithm (denoted as BiD + CV in Table IV) outperforms in both aspects.

a) Loop closure accuracy: First, we calculate the Euclidean distance from the corresponding poses of the loop closure. For homogenous inter-robot loop closure, the Euclidean distance is computed in 3-D space. Moreover, in the case of heterogeneous loop closure characterized by the difference in height space, only the Euclidean distance in the XY plane is considered. A loop closure qualifies as true if this distance falls below the predefined threshold λ. Second, loop closure accuracy is the ratio of true loop closures to the total detected. Let L_{FD} denote the total number of loop closures obtained by FD, where L_{FD}^{t} represents the true loop closures. The loop closure accuracy F_{FD} is calculated as

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TABLE IV
COMPARISON OF LOOP ACCURACY BETWEEN DIFFERENT LOOP CLOSURE DETECTION METHODS AS THE LOOP DISTANCE THRESHOLD λ VARIES

	Forward Detection	Reverse Detection	Bidirectional Verification	Consistency Verification	KITTI		DCSI-UAV		DCSI-ATCM	
					1m	2m	0.5m	1m	0.5m	1m
FD	✓				0.11	0.28	0.23	0.84	0.32	0.68
RD		✓			0.10	0.28	0.21	0.84	0.31	0.72
BiD	✓	✓	✓		0.32	0.77	0.26	0.89	0.41	0.80
FD+CV	✓			✓	0.37	0.90	0.34	0.91	0.32	0.69
BiD+CV	✓	✓	✓	✓	**0.43**	**0.99**	**0.42**	**0.91**	**0.42**	**0.81**

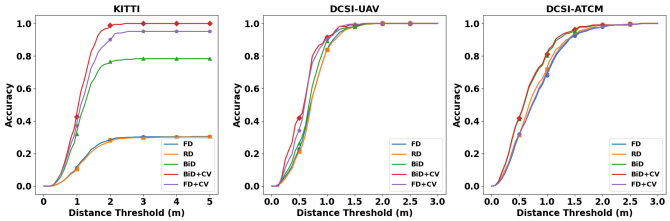


Fig. 10. Comparative curve of loop accuracy between different loop closure detection methods. (a) KITTI. (b) DCSI-UAV. (c) DCSI-ATCM.

TABLE V
COMPARISON OF TRAJECTORY PRECISION BETWEEN DIFFERENT LOOP CLOSURE DETECTION METHODS

Dataset	FD+CV/BiD+CV			
	APE		RPE	
	RMSE	median	RMSE	median
KITTI	119.54/**1.07**	109.10/**0.85**	7.18/**0.08**	0.33/0.33
DCSI-UAV	7.32/**0.32**	5.88/**0.16**	1.14/**0.35**	0.05/0.05

$F_{\mathrm{FD}} = (L^t_{\mathrm{FD}}/L_{\mathrm{FD}})$. For the extensive KITTI dataset, the threshold λ is specified as $\lambda \in [0, 5]$ m, and for the DCSI–ATCM dataset, λ is set to $\lambda \in [0, 3]$ m. In the accuracy evaluation of this section, it includes all inter-robot loop closures, without distinguishing the heterogeneous or homogeneous cases.

In Table IV, BiD + CV demonstrates significantly higher loop closure accuracy compared to other modules. In the DCSI-ATCM dataset, with $\lambda = 0.5$, it achieves a 31.25% improvement over FD + CV. Fig. 10 illustrates the evolution of loop closure accuracy across stages in different datasets, showing continuous improvement as additional modules are integrated. Our BiD + CV algorithm consistently achieves the highest precision, validating the effectiveness of each module in the BLD algorithm.

b) Trajectory precision: To evaluate the benefits of the BLD algorithm, we compare global trajectory precision using unidirectional and bidirectional loop closure detection in Table V. In both the KITTI dataset and the DCSI-UAV dataset, BLD demonstrates significantly higher trajectory precision compared to traditional unidirectional loop closure detection. The reason is that the unidirectional loop closure detection incorporates more incorrect loop closures into the back-end PGO, reducing trajectory precision. In addition, the experiments also validate the necessity of improving loop closure accuracy.

TABLE VI
COMPARISON OF LOOP NUMBER AND ACCURACY AS THE LOOP DISTANCE THRESHOLD λ VARIES

Method	UGV1-UAV / UGV2-UAV	
	Loop number	Loop accuracy ($\lambda = 1.5/2m$)
Scan context++	0	
Ours	209 / 210	(0.89/0.96) / (0.90/0.97)

2) Cross-View Global Loop Descriptor: The CGLD designed for heterogeneous collaborative mapping is evaluated on the DCSI–ATCM dataset. Since the proposed CGLD is based on geometric features, we select scan context++ [26], a method of the same category for comparative analysis. The parameters of scan context++ are consistent with those of our method on the DCSI-ATCM dataset, including the initial LiDAR height above ground for both UGV and UAV platforms. The proposed CGLD algorithm's effectiveness is demonstrated from two key aspects: loop closure quantity and accuracy. We compare heterogeneous inter-robot loop closures obtained by forward detection. As shown in Table VI, the traditional scan context++ method lacks viewpoint correction capabilities, resulting in ineffective detection. In contrast, the CGLD algorithm shows superior performance, achieving an accuracy of 0.97 with a loop distance threshold of $\lambda = 2$ m. In addition, the method identified a total of 419 heterogeneous inter-robot loop closures, validating CGLD's cross-viewpoint capabilities and its effectiveness in addressing the challenges of heterogeneous inter-robot loop closure detection.

3) Two-Stage GPGO: We conducted ablation analyses on the KITTI and DCSI-UAV datasets, comparing different stages of GPGO, including a one-stage method with only the first stage of coordinate transformation and the complete two-stage method. Table VII shows that for the KITTI dataset using the SCALOAM method, the RMSE of the two-stage APE is 1.07, which represents a 23.0% improvement over the one-stage method. The improvements based on KISS-ICP and CT-ICP methods are more obvious, reaching 65.72% and 69.46%, respectively. In the DCSI-UAV dataset, with the SCALOAM method, the RMSE of the two-stage APE is 0.32, indicating a 42.85% improvement than the one-stage method.

D. Discussion

In this article, the proposed ATCM method integrates CGLD for heterogeneous inter-robot loop closure detection, enhancing accuracy through the BLD algorithm and employing a two-stage GPGO approach for fusing diverse multisource information. This distinguishes it from

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TABLE VII
COMPARISON OF APE AND RPE BETWEEN ONE- AND TWO-STAGE GPGOS BASED ON DIFFERENT FRONT-END ALGORITHMS

		KITTI				DCSI-UAV			
		APE		RPE		APE		RPE	
		RMSE	median	RMSE	median	RMSE	median	RMSE	median
SCALOAM	one-stage	1.39	1.22	0.18	0.04	0.56	0.29	0.38	0.05
	two-stage	**1.07**	**0.85**	**0.08**	**0.03**	**0.32**	**0.16**	**0.35**	0.05
KISS-ICP	one-stage	4.96	3.33	0.7	0.04	failed			
	two-stage	**1.7**	**1.46**	**0.08**	0.04				
CT-ICP	one-stage	7.63	6.05	0.41	0.04	failed			
	two-stage	**2.33**	**1.82**	**0.09**	0.04				

methodologies employed for homogeneous inter-robot loop closure detection, such as those based on scan context [20] and iris [21]. The front end adopts LiDAR-only odometry instead of multisensor fusion because it is decoupled with the back end in this article. It only needs to provide the locally optimal poses and the corresponding keyframes to the back end, ensuring compatibility with various algorithms. To simplify the validation of our method's effectiveness, we adopt the LiDAR-only odometry method.

However, there are still some limitations in this article. The introduced CGLD relies solely on geometric height information and lacks the incorporation of high-level semantics for loop closure. Beyond geometric height information, future work will explore the integration of lightweight semantic segmentation networks to obtain static semantic classes (e.g., buildings and sidewalks) and incorporate per-bin semantic distributions into descriptor similarity metrics. Integrating semantic information would require more computational resources and may face challenges in scenarios with degraded geometric features. In addition, significant viewpoint height differences between heterogeneous platforms may reduce the number of detected loop closures. Furthermore, to enhance mapping and trajectory precision, we propose a bidirectional loop closure detection method, which improves interloop constraint accuracy but may reduce loop closure quantity. Our experiments show that loop closure accuracy significantly affects trajectory precision more than the number of loop closures. In the future, our aim is to enhance the CGLD by integrating more advanced features to address complex environments. We will also explore more efficient multi-robot loop closure retrieval methods.

V. CONCLUSION

In this article, we introduce ATCM, a C-SLAM method designed for aerial and terrestrial robots that efficiently fuse heterogeneous cross-view information. It includes a front-end compatible with various single-robot SLAM algorithms. A novel global point cloud descriptor, CGLD, is introduced to uniformly map cross-view information into the height domain, enabling cross-view loop closure detection. In addition, the BLD algorithm improves loop detection precision through further validation. The effectiveness of the proposed ATCM is demonstrated on diverse datasets, showing a 21.95% improvement in trajectory accuracy and a 17.00% enhancement in map precision compared to state-of-the-art LiDAR-based odometry. CGLD achieves an accuracy of 0.97 in detecting heterogeneous inter-robot loop closures. Moreover, BLD methods show a notable 31.2% improvement in accuracy over traditional unidirectional methods. Future work will explore incorporating high-level information for describing point clouds in complex environments.

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